



**DAVANGERE UNIVERSITY
DON BOSCO COLLEGE
CHITRADURGA-577501**



**PROJECT REPORT ON
FAKE NEWS DETECTION USING AI**

A Project submitted for the partial fulfilment of the requirements for qualifying

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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DON BOSCO COLLEGE
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2025-2026

CERTIFICATE

This is to certify that the project entitled "**FAKE NEWS DETECTION USING AI**" carried out by **KETHAN C (U13DO23S0075), THEM GUOT KONPINY (U13DO23S0127), P.G VINUTHA (U13DO23S0080), RESHMABHANU H (U13DO23S0085)** student of **DON BOSCO COLLEGE, CHITRADURGA**. In partial fulfilment of the requirements for the award of **BACHELOR OF COMPUTER SCIENCE (BSC)** of the **DAVANGERE UNIVERSITY, CHITRADURGA** during the year 2025-2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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External Examiner

Date:

Internal Examiner

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DECLARATION

We hereby declare that the dissertation work entitled " **FAKE NEWS DETECTION USING AI**" which is being submitted by us in the partial fulfilment for the award of the degree of Bachelor Of Computer science from **DAVANGERE UNIVERSITY**, Tholahunase Davangere is an authentic record of our own independent work carried out by me during the academic year 2025-2026, under the valuable guidance of Lecturer miss **Sreeshma Shibu BSC., MSC**, Department of Computer Science of **DON BOSCO COLLEGE CHITRADURGA** .The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree and due acknowledgement has been made in the text and all other material used.

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ABSTRACT

The rapid growth of digital media platforms and online news sources has significantly increased the spread of misinformation and fake news. This has created serious social, political, and economic challenges. The **Fake News Detection System** is developed to automatically classify news content as *real* or *fake* using advanced Machine Learning (ML) and Natural Language Processing (NLP) techniques.

The system allows users to input either raw text or a news URL. The input data is processed through several stages including data cleaning, tokenization, stop-word removal, and feature extraction. Techniques such as **TF-IDF (Term Frequency–Inverse Document Frequency)** and classification algorithms like **Logistic Regression / Naïve Bayes / Support Vector Machine (SVM)** are used to train the model.

The backend of the application is developed using **Flask**, while the frontend provides an interactive and modern user interface with features like:

- News input panel (text + URL)
- Prediction result with confidence score
- History tracking of previous results
- User authentication system
- Admin controls (optional)

Additionally, the system integrates AI-based validation and live web search to enhance prediction accuracy. The model is optimized to run efficiently on low-resource systems (such as Intel i5 with 8GB RAM), making it suitable for practical deployment.

The primary objective of this project is to minimize the spread of fake news by providing users with a reliable and easy-to-use verification tool. This system can be further extended for use in social media monitoring, journalism, and digital content verification.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Fake news detection is the study and application of techniques to identify false, misleading or intentionally fabricated information that is presented as real news. It has become an important topic with the rise of digital media especially in fields like AI, data science and natural processing.

In today's digital world, news spread rapidly through social media platform, websites and messaging apps. While this allows quick access to information, it also makes it easy for false or misleading content known as fake news to spread widely. It is often created to influence public opinion, gain financial benefits through clicks or create confusion.

Purpose for fake news detection

- To identify and filter misinformation
- To protect users from being misled
- To maintain trust in news and media system
- To support informed decision making

1.2 Background of fake news detection

Fake news is not a modern phenomenon. It has existed for centuries in forms such as

- Political propaganda
- Rumors and misinformation
- Biased or exaggerated reporting

With the development of mass media, information reached larger audiences, government and organization used media for influences.

The background of fake news lies in its long history but it has become a major issue in the digital age due to rapid information sharing. Understanding its origin and impact is essential for developing effective solutions to combat misinformation.

1.3 Importances of fake news detection

1.3.1 Prevents spread of misinformation

- Identify false or misleading content
- Stop it from spreading further
- Reduce confusion among the public

1.3.2 Protects public awareness and decision making

- Access to accurate information
-

- Better personal and societal
- Reduced influence of false narratives
- Prevent propaganda and misinformation campaigns

1.3.3 Safeguards democracy

- Promote fair and transparent elections
- Maintain trust in democratic system

1.3.4 Maintain media in trust

- Help distinguish credible sources
- Support reliable news organization
- Restore public trust in media

1.3.5 Reduce social harm

- Panic during crisis
- Social unrest and conflicts
- Damage to reputation

1.4 Challenges in fake news detection

Detecting fake news is a complex task due to the dynamic and deceptive nature of misinformation. Even with advancements in artificial intelligence, machine learning and natural languages processing, several challenges remain:

1.4.1 Rapid spread on social media

- Fake news spreads faster than real news especially on platforms like twitter and Facebook
- Detection system often lags behind viral content

1.4.2 Tactics Evolving

- Creators of fake new constantly adapt to bypass detection systems
- Use of clickbait headlines, deepfakes and coordinated bot networks makes detection harder

1.4.3 Lack of high-quality labeled data

Machine learning models need large datasets labeled as “fake” or “real”

Such datasets are:

Expensive to curate

Often biased

- Limited

1.4.4 Multimodal content

- Fake news is no longer just a text, it includes images, videos and audio
- Detecting manipulated media such as deepfakes is especially difficult

1.5 How it works

1.5.1 Data collection

- News articles, social media posts and metadata are gathered from platforms like Twitter and Facebook
- Fact-checking sources provide labeled examples of real and fake news

1.5.2 Preprocess

- Remove stop words (e.g. “the”, “is”)
- Convert text to lowercase
- Tokenization (splitting into words)

This prepares raw data for analysis

1.5.3 Feature extraction

- The system converts text into measurable features

1.5.4 Model training

Machine learning models are trained using labeled data such as

- Logistic regression
- Decision trees
- Neural approaches use deep learning

1.5.5 Classification

When new content appears, the trained model predicts:

- Fake or real
- Sometimes a probability score (e.g. 85% likely fake)

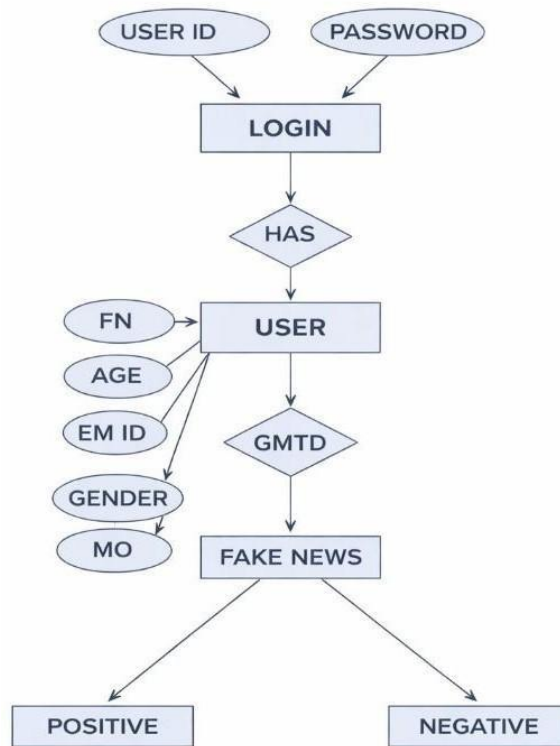
1.5.6 Fact checking integration

- System compares claims with verified databases:
- Knowledge graphs
- Trusted sources
- Some platforms integrate human fact checkers

1.5.7 Multimodal analysis

- Detects fake content beyond text:
- Image → reverse image search
- Videos → deepfake detection
- Combines multiple signals for better accuracy

1.6 ER diagram on how fake news detection works



CHAPTER 2

SYSTEM STUDY

2.1 Existing system

Fake news detection has become an important area of research and application especially with the rise of misinformation on social media platforms and news outlets. Existing systems for detecting fake news generally rely on mix of computational techniques from natural language processing (NLP), machine learning (ML) and deep learning (DL)

Here are the key components of existing system for fake news detection

2.1.1 Data collection

- **Sources based approach:** One of the simplest methods involves looking at the sources of news, reputable sources are likely more to report accurate information, while unreliable or biased sources may propagate fake news
- **Social media monitoring:** Fake news often spread rapidly on platforms like twitter Facebook, Reddit etc. tools often scrape these to collect news articles, social media posts, comments etc. for analysis.

2.1.2 Data preprocessing

- Cleaning text (removing stop words, punctuation)
- Tokenization, stemming and lemmatization
- Handling missing or noisy data

2.1.3 Features Extraction: Transform raw text into numerical representations

- Linguistic features: = sentiment, syntax, readability
- Metadata feature: = author credibility, public time
- Social features: = shares, likes, propagation pattern

2.1.4 social context analysis

- Examines how information spreads across networks
- Detect bots, coordinated campaigns and echo chambers

2.1.5 Classification/ prediction: The trained model classifies content as:

- Fake
- Real
- Sometimes also “partially true” or “misleading”

2.2 Limitation of the existing system

2.2.1 Dependence on labeled data

Most systems rely on supervised learning, which needs large datasets of labeled “real” vs “fake” news.

- High quality labeled data is scarce and expensive to create.
- labels can be subjective (what counts as “fake” isn’t always clear).
- models struggle when applied to new domains or topics.

2.2.2 Poor generalization

Models trained on one dataset often fail on new or unseen data

- Fake news styles evolve quickly
- Languages, tone and formats change across platforms
- Models may overfit to specific pattern instead of learning deeper meaning

2.2.3 limited fact-checking ability

- detection systems don’t truly “understand” facts
- they rely on patterns rather than verifying claims
- real-time fact-checking against reliable sources is difficult
- knowledge bases may be outdated or incomplete

2.2.4 Real-time detection limitation

- Processing large volumes of news quickly is difficult
- Delays in detection allow fake news to spread widely before being flagged
- Scalability remain a concern

2.3 proposed system

The system uses a hybrid approach combining natural language machine learning and fact-checking to identify fake new. a web-based Fake News Detection application acts as a blueprint that integrates machine learning logic with a functional user interface. It transitions the project from a static data science script into a dynamic, interactive tool.

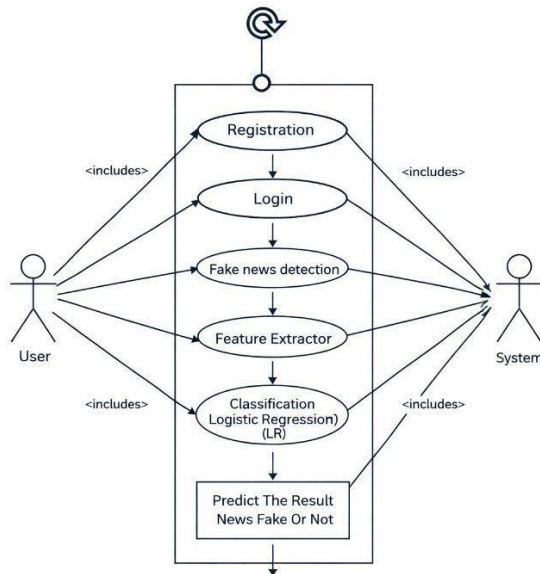
2.3.1 Key features

- Source credibility
- Author credibility
- Content analysis
- Fact-checking
- Network

CHAPTER 3

SYSTEM DESIGN

3.1 USE CASE DIAGRAM



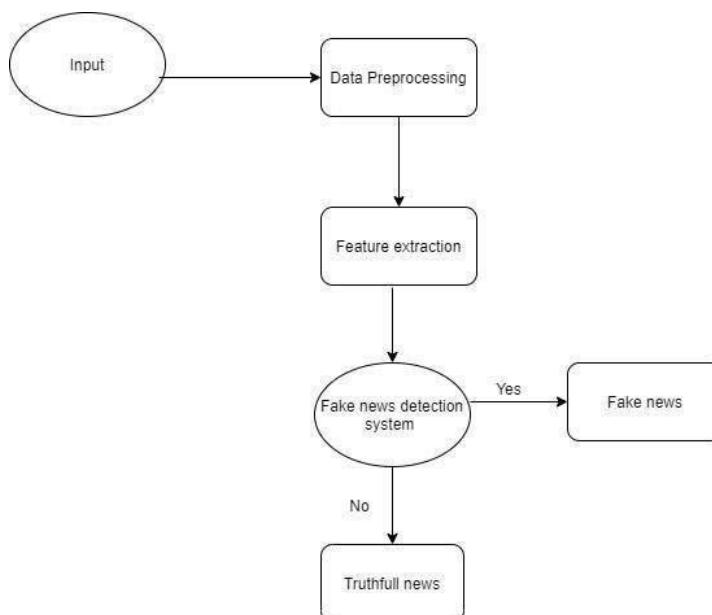
3.2 DESCRIPTION OF DATA FLOW

Data flow pipeline is shown at the table

Stage	Process description	Data output
1.ingestion	Collecting raw content from social APIS, RSS feeds or web scrapers.	Raw data: headlines, body text, and metadata (author, timestamp).
2.cleaning	Removing “noise” such as HTML tags, emojis and irrelevant punctuation.	Cleaned text: standardized, plain-text strings.
3.processing	Tokenization, removing stops words and performing	Processed tokens: individual keywords or word units ready for analysis.

	lemmatization (reducing words to roots).	
4.vectorization	Converting text into numerical values using methods like TF-IDT or word embeddings.	Features vectors: mathematical representations of the text's meaning.
5.evidence retrieval	Querying external databases or “Gold standard” news sources to find corroborating facts.	Contextual evidence: supporting or contradicting snippets from verified sources.
6.classification	Feeding vectors into a machine learning model to calculate patterns	Probability score: A numerical value indicating the likelihood of truth.
7.final output	Applying a threshold to the score to assign a human-readable tags.	Classification label: “Real,” “fake,” “Misleading,” or “satire”.

3.3 DATA FLOW DIAGRAM OF THE SYSTEM



3.4 ALGORITHMS USED

3.4.1 HTML

Hypertext markup language is the standard backbone of every website on the internet. It isn't a programming language in the traditional sense (it doesn't handle logic or math); instead, it is a markup language used to structure content on a page.

How It Works: The Building Blocks

HTML uses **Tags** to label content. Most tags come in pairs: an opening tag `<tag>` and a closing tag `</tag>`.

- **Elements:** The combination of the tags and the content inside them.
- **Attributes:** Extra information provided inside the opening tag (like a link's destination).
- To keep things organized, HTML uses specific tags for different types of content:

Category	Tags	Purposes
Structure	<code><html></code> , <code><body></code> , <code></head></code>	Define the overall document layout
Text	<code><h1></code> , to <code><h6></code> , <code><p></code> , <code></code>	Creates headings, paragraphs and bold text.
Links	<code><a></code>	Connects one page to another using the href attribute.
Media	<code></code> , <code><video></code> , <code><audio></code>	Embeds visual or auditory content.
Lists	<code></code> , <code></code> , <code></code>	Creates bulleted or numbered list

3.4.2 INTRODUCTION TO PYTHON PROGRAMMING

Python is a high-level, interpreted programming language that has gained immense popularity due to its simplicity, readability, and versatility. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability with its use of significant indentation. This makes Python particularly attractive for beginners, as its syntax is clear and straightforward. Python is dynamically typed, meaning that variable types are determined at runtime, allowing for greater flexibility in coding. It boasts an extensive standard library that supports a wide array of programming tasks, from web development and data analysis to artificial intelligence and scientific computing. Python is also cross-platform, enabling code to run on various operating systems such as Windows, macOS, and Linux. The language has a large and active community, contributing to a rich ecosystem of libraries and frameworks like Django for web development, Pandas for data manipulation, and TensorFlow for machine learning. These features make Python a powerful tool for both simple scripting and complex application development, fostering a supportive environment for developers of all levels.

HISTORY OF PYTHON

Python, created by Guido van Rossum, was first released in 1991. Its development began in the late 1980s at Centrum Wiskunde & Informatica (CWI) in the Netherlands, where Van Rossum was inspired by the ABC language, which emphasized simplicity and readability. Python was designed to be easy to understand and use, with a focus on code readability and a syntax that allowed programmers to express concepts in fewer lines of code compared to other languages like C++ or JavaScript.

In October 2000, Python 2.0 was released, introducing list comprehensions, garbage collection, and the first version of the `print` statement as a function. Python 2.x versions continued to improve the language, but certain design flaws and the need for more modern features led to the development of Python 3.

Python 3.0, released in December 2008, was a major overhaul aimed at removing redundant features and fixing inconsistencies in the language design. This new version was not backward-compatible with Python 2.x, which initially caused some resistance in the community. However, Python 3 introduced several new features, such as a more consistent syntax, improvements to the standard library, and enhancements in performance.

Over the years, Python 3.x has become the standard, with Python 2 reaching its end of life in

January 2020. The transition from Python 2 to Python 3 was supported by the community through tools like `2to3`, which helped automate code conversion.

Today, Python continues to grow in popularity due to its simplicity and versatility. It is widely used in web development, scientific computing, data analysis, artificial intelligence, and more. The language's active community, extensive libraries, and frameworks contribute to its ongoing evolution and success, ensuring that Python remains a top choice for developers around the world.

APPLICATION OF PYTHON

Python is renowned for its versatility and wide range of applications across various domains. Here are some of the key areas where Python is extensively used:

- **Web Development:** Python provides robust frameworks like Django, Flask, and Pyramid that streamline web development by offering powerful tools and libraries for building scalable, secure, and maintainable web applications. Django, for instance, is known for its "batteries-included" philosophy, offering built-in features like authentication, database management, and an admin interface.
- **Data Science and Analytics:** Python has become the go-to language for data scientists due to its simplicity and the extensive range of libraries available for data analysis, visualization, and machine learning. Libraries such as Pandas, NumPy, and Matplotlib facilitate data manipulation and visualization, while Scikit-learn and TensorFlow are widely used for machine learning and deep learning applications.
- **Artificial Intelligence and Machine Learning:** Python's simplicity and the powerful libraries it offers make it ideal for AI and ML projects. TensorFlow, Keras, and PyTorch are popular frameworks that enable the development of complex neural networks and deep learning models. These libraries provide pre-built models and tools for tasks like natural language processing, image recognition, and predictive analytics.
- **Automation and Scripting:** Python's ease of use and readability make it a perfect choice for writing scripts to automate repetitive tasks. It is commonly used for automating system administration tasks, data scraping from websites, and automating software testing processes. Libraries like Selenium and BeautifulSoup enhance its capabilities in web scraping and browser automation.
- **Game Development:** Python is used in game development for prototyping and developing simple games. Libraries like Pygame provide functionality for game development,

including handling graphics, sound, and user input.

- **Education:** Python's simplicity and readability make it an excellent language for teaching programming and computer science concepts. It is often the first language taught in introductory programming courses around the world.
- **Software Development:** Python is used for developing a variety of software applications, from desktop applications to complex enterprise systems. Its integration capabilities with other languages and tools make it a versatile choice for software development projects.

Key Characteristics

- Python stands out because of a few fundamental design choices:
- **Interpreted:** You don't need to compile your code into machine language before running it. An "interpreter" reads and executes the code line-by-line, which makes debugging much faster.
- **Dynamically Typed:** You don't have to explicitly declare whether a variable is a number, a piece of text, or a list; Python figures it out automatically.
- **Indentation Matters:** Unlike other languages that use curly braces { } to group code, Python uses whitespace (indentation). This forces developers to write clean, organized code.
- **Importances of python**
- Python is the primary language for Artificial Intelligence, Machine Learning, and Data Science.
- Python's syntax is clean and mirrors English.
- Python is the king of automation. It is used globally to automate "boring" or repetitive tasks—such as organizing files.
- With millions of developers worldwide, Python has an unparalleled support system.

3.4.3 JavaScript

JavaScript (JS) is the programming language of the web. While HTML provides the structure and CSS provides the style, JavaScript provides the interactivity. It is what allows a webpage to respond to user actions without reloading the entire page.

Use of JavaScript

3.4.3.1 Front-End Web Development

This is JavaScript's "home." It is used to create everything interactive you see on a website.

- **Dynamic Content:** Updating news feeds, showing notifications, or loading new emails without refreshing the page.
- **UI Components:** Building complex dropdown menus, image carousels, and interactive maps.

3.4.3.2 Back-End (Server-Side) Development

Through **Node.js**, JavaScript runs on servers just like Python or Java.

- **Web Servers:** Handling requests from millions of users simultaneously (used by Netflix, LinkedIn, and Uber).
- **APIs:** Building the "hooks" that allow different software programs to talk to each other.

3.4.3.3 Development Mobile App

Instead of learning Swift (for iOS) or Kotlin (for Android), developers use JavaScript to build for both at the same time.

- **React Native:** A popular framework that lets you write JS code that renders as a truly "native" app on your phone.
- **Progressive Web Apps (PWAs):** Websites that can be installed on a phone and work offline, just like a regular app

3.5 Data collection modulo

"Data Collection Module" is the critical first stage of the pipeline. Its job is to gather raw information from various sources before it is processed for veracity.

Because "fake news" isn't just text—it's a combination of social patterns, timing, and intent—this module must be highly sophisticated.

3.5.1 Primary Sources of Data

A robust data collection module typically pulls from three main areas:

- **News Content:** The actual headlines and body text from news sites, blogs, and mainstream media outlets.
- **Social Context:** Engagement data from platforms like X (Twitter), Facebook, or Reddit. This includes likes, shares, retweets, and comments.
- **User Profiles:** Information about the people sharing the news, such as account age, verification status, and past posting history.

3.5.2 The Mechanics of Collection

The module uses different technical tools depending on the source:

- **Web Scraping:** Using libraries (like BeautifulSoup or Scrapy) to "read" news websites and extract text, authors, and publication dates.
- **APIs:** Leveraging official platform interfaces (like the X API) to pull real-time streams of posts and their associated metadata.
- **Crowdsourcing:** Some modules pull data from fact-checking sites (like Snopes or PolitiFact) to build "ground truth" datasets for training.

3.6 Features extraction

Feature Extraction is a critical step in the machine learning and data science pipeline. It is the process of transforming raw, messy data into a set of numerical characteristics (features) that an algorithm can actually understand and process.

3.6.1 Benefits of Good Feature Extraction

- **Accuracy:** -Better features lead to better predictions.
- **Efficiency:** - It reduces the amount of data the computer has to process, making the model faster.
- **Generalization:** -It helps the model focus on the *patterns* rather than the specific details of one training set.

3.7 prediction interface

prediction interface (also known as a Predictive User Interface or PUI) is no longer just a static dashboard. It is a dynamic system that uses machine learning to anticipate a user's next move, surface relevant information before it is requested, and adapt its layout in real-time

3.7.1 Core Mechanisms of Predictive Interfaces

Modern prediction interfaces operate on a spectrum of automation, balancing helpfulness with user control:

- **Behavioural Pattern Recognition:** Analysing sequences (e.g., if you usually open "Reports" after "Dashboard," the "Reports" button might pulse or move to a more prominent position).
- **Context-Aware Synthesis:** Real-time adjustments based on Temporal (time of day), Spatial (location), and Device context.
- **Confidence Scoring:** The system calculates a probability for its prediction.

3.7.2 Design Patterns

To avoid being intrusive (the "Clippy" effect), designers use specific patterns to integrate predictions seamlessly:

Pattern	Description	Example
Anticipatory	Navigation that highlights the most likely next step.	A "Next: Export to PDF" button appearing after data analysis.
Breadcrumbs	Low-risk actions happen automatically; high-risk ones ask first.	Auto-saving a draft vs. asking before sending a mass email.
Progressive Confirmation		

3.7.3 Technical Architecture

Building these interfaces requires a robust "Prediction Stack" to ensure the UI remains snappy:

- **Client-Side Heuristics:** Simple, fast rules handled on your device for immediate feedback (e.g., hover states).

- **Edge ML Models:** Critical predictions processed near the user to reduce latency (e.g., gesture recognition).
- **Cloud Inference:** Complex, heavy lifting that analysis massive datasets to provide deep personalization (e.g., long-term trend forecasting).

3.7.4 The "Silent Operator" Ethics

- **Attribution:** Good interfaces briefly explain *why* a prediction was made.
- **Frictionless Override:** If the prediction is wrong, the user must be able to dismiss it with a single tap. This feedback loop is then used to retrain the model.
- **Privacy Boundaries:** Interfaces are designed to never "predict" sensitive or protected personal information (like health status or financial debt) to avoid making users feel shrivelled.

3.8 Model training and evaluation

3.8.1 Data Collection & Preprocessing

Modern datasets prioritize diversity across domains (Politics, Health, Tech) and geographic origins to prevent "overfitting" to a specific event.

- **Standard Benchmarks:** ISOT (full articles), LIAR (short statements), and faked it (multimodal).
- **Emerging Datasets:** Look for the Global Fake News Research Dataset, which includes engineered features like "Clickbait Score," "Source Trust Score," and "Emotional Word Ratio."
- **Preprocessing Pipeline:**
 - **Cleaning:** Removing URLs, HTML tags, and non-alphanumeric noise.
 - **Normalization:** Lemmatization (reducing words to their roots) is preferred over stemming to maintain semantic meaning.
 - **Feature Engineering:** Extracting TF-IDF (for keyword frequency) and 768-dimensional embeddings (using models like RoBERTa or BERT) for deep semantic understanding.

3.8.2 Model Training Strategies

The industry uses a two-tier approach:

- **Baseline:** Classical Machine Learning
For high-speed, interpretable tasks, Random Forest and Logistic Regression remain dominant.
- **Performance:** These often achieve 98–99% accuracy on structured datasets like ISOT but struggle with "micro-statements" like tweets.
- **Ensemble Methods:** A "Soft-Voting" ensemble (combining Random Forest, SVM, and Logistic Regression) is used to stabilize predictions by averaging probabilities.
- **B. SOTA:** Deep Learning & Multimodal
- **Transformer Architectures:** Models like RoBERTa or GPT-4-based detectors identify stylistic nuances (e.g., overly sensational tone) that simple keywords miss.
- **Multimodal Integration:** Using ViLBERT (Vision-and-Language BERT) to check if an image matches the context of the text. Mismatches are a primary indicator of "out-of-context" fake news.

3.8.3 Evaluation Metrics

Relying solely on "Accuracy" is a common pitfall in fake news detection due to class imbalance (e.g., more real news than fake).

Metric	Why it Matters
Precision	Minimizes "False Positives" (labelling real news as fake), which is vital for maintaining user trust in platforms.
Recall	Minimizes "False Negatives" (missing actual fake news), crucial for safety and misinformation containment.
F1-Score	The harmonic mean of Precision and Recall; the gold standard for balanced performance.
AUC-ROC	Measures how well the model distinguishes between classes across different thresholds
Confidence Score	Modern interfaces now require models to output a probability (e.g., "85% likely to be false") rather than a binary label.

CHAPTER 4

METHODOLOGY

4.1 Waterfall model

Waterfall Model is the oldest and most traditional methodology used in software development and project management. It follows a linear, sequential approach, meaning each phase must be completed 100% before the next one begins.

Just like a physical waterfall, the progress flows steadily downwards through the various stages and cannot easily flow back "up."

4.2 The Six Stages of Waterfall

The workflow follows these specific steps:

- **Requirements Analysis:**

All possible requirements of the system to be developed are captured in a document. This is the most critical stage, as any misunderstanding here can ruin the entire project.

- **System Design:**

The requirements from the first phase are studied and the system design is prepared. This helps in specifying hardware and system requirements and defining the overall system architecture.

- **Implementation (Coding):**

The system is first developed in small programs called units, which are integrated in the next phase.

- **Integration and Testing:**

All the units developed in the implementation phase are integrated into a system and tested for any faults or failures.

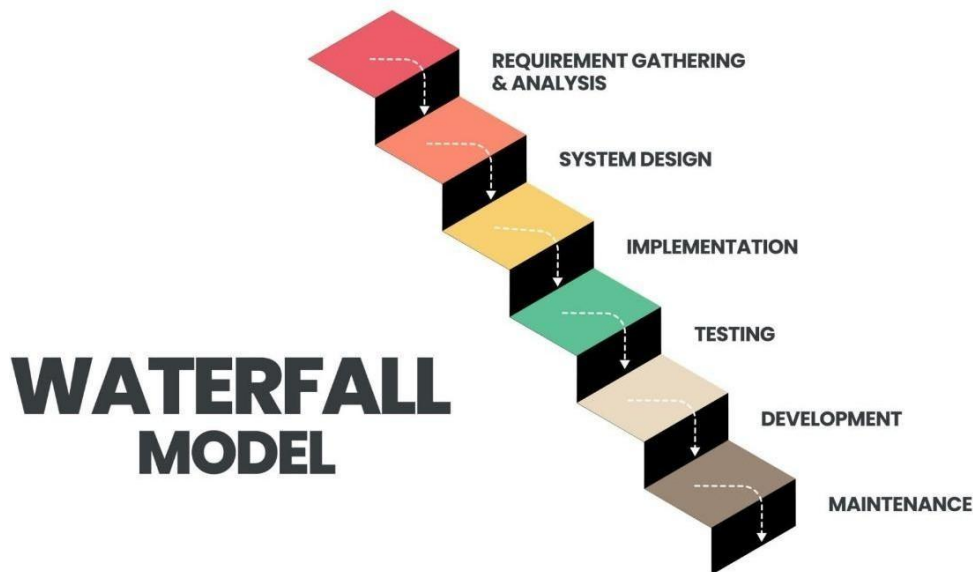
- **Deployment:**

Once the functional and non-functional testing is done, the product is deployed in the customer environment or released into the market.

- **Maintenance:**

Problems may come up in the client environment. To fix those issues, patches are released. Also, to enhance the product, better versions are released.

4.3 Diagram for the waterfall stages



4.4 When to Use Waterfall

Waterfall is not "outdated," but it is specialized. It works best when;

- The project is small and simple.
- The requirements are very clear and fixed (they won't change).
- The product definition is stable.
- The technology is understood and not dynamic.
- There are no ambiguous requirements.
- Ample resources with required expertise are available freely.

4.5 Advantages vs. Disadvantages

Advantages

- Simple and easy to understand
- It provides high visibility for stakeholders. Project managers can easily track progress against the timeline
- Fixed and clear; prevents "scope creep."
- Highly structured and methodical.
- Easy to estimate upfront for contracts.

Disadvantages

- If requirements are wrong, the whole project fails

- Happens too late in the process to catch fundamental flaws.
- Very expensive to fix mistakes found late in the game.
- Rigid; does not account for the "human element" of changing minds.

4.6 Primary Goals

- **Establishing a Stable "Ground Truth"**

To finalize the labelling criteria (e.g., using only verified Snopes ratings) during the Requirements phase so the model isn't trained on shifting definitions.

- **Rigorous Feature Architecture**

To complete a comprehensive **System Design** that accounts for all linguistic variables, ensuring the model's logic is sound before the computationally expensive training phase begins.

- **Minimizing "Data Leakage"**

Use the sequential nature of Waterfall to strictly separate the Data Acquisition phase from the Testing phase, ensuring the model is evaluated on truly "unseen" news stories.

- **High Reproducibility for Peer Review**

To produce exhaustive documentation at every stage (Data Cleaning logs, Hyperparameter settings, etc.) so that any other researcher can follow your "waterfall" and reach the same accuracy.

Why it matters for Fake News

- Ensures that "Real" vs "Fake" labels are audited and locked in before the model starts "learning" biases.
- Training deep learning models (like BERT) is expensive. Waterfall ensures the design is perfect *before* you spend money/time on GPU training.
- If a model wrongly flags a news story, the Waterfall documentation allows you to trace the error back to a specific design or requirement flaw.
- It enforces a clean pipeline: You cannot analyse data until you have cleaned it and defined it

4.7 Key Characteristics:

- It moves from top to bottom like a physical waterfall.
- Every stage requires a formal "sign-off" and extensive documentation.
- Because the scope is fixed at the start, it is easier to calculate budgets and timelines
- It is difficult and expensive to change requirements once the project has moved into the coding or testing phases.

CHAPTER 5

TESTING

5.1 Testing

Testing is the systematic evaluation of a detection model to ensure it can distinguish between authentic journalism and various forms of misinformation. This involves verifying not just the text, but also the metadata (like watermarking) and the propagation patterns (how the news spreads).

Testing ensures the reliability and accuracy of the fake news detection model. Unit testing checks individual components, while integration testing validates the interaction between data preprocessing and model prediction. The system is evaluated on the test datasets and performance metrics, precision, recall and F1 score are calculated.

Types of Testing (Detection Strategies):

To ensure a system is robust, developers use several "tracks" of testing:

- **Stylistic & Linguistic Testing:** Analyses the "writing DNA." Fake news often uses high Emotional Word Ratios, clickbait headlines, and inflammatory sentiment.
- **Fact-Checking Verification:** The system cross-references claim against trusted "Knowledge Graphs" and real-time databases (like the Gemini 1.5 Flash API or verified fact-checking sites)
- **Multi-modal Testing:** This tests if the system can catch GAN-constructed footage (AI-generated video) or altered images that accompany the text.
- **Adversarial Testing:** Pitting the detection system against "LLM-groomed" content—fake news specifically written by AI to mimic the style of reputable sources like the *New York Times*.
- **Social Network Analysis:** Testing the system's ability to identify bot networks or suspicious propagation "bursts" on social media.

5.2 Importance of Testing

Testing is critical because the cost of failure is high:

- **Public Safety:** Misinformation during health crises or natural disasters can lead to loss of life.
- **Democratic Integrity:** Detecting deepfakes and fabricated documents is vital during election cycles to prevent voter manipulation.
- **Algorithmic Trust:** If a system flags too many real stories (False Positives), users will stop trusting the tool entirely.

- **Legal Compliance:** Under 2026 regulations (like the updated IT Rules), platforms are increasingly liable for failing to flag provably synthetic content.

5.3 Test Strategy

The strategy follows a Multi-Layered Verification model, ensuring the system can handle both the content (what is said) and the context (how it spreads).

- **The "Three-Pillar" Approach**

- **Semantic Analysis:** Testing the system's ability to detect linguistic cues—like sensationalism, emotional manipulation (sentiment scoring), and structural inconsistencies.
- **External Fact-Checking:** Verifying claims against real-time "Knowledge Graphs" and reputable news databases (e.g., Reuters, AP).
- **Social Context Tracking:** Testing the model's ability to analyse metadata, such as account age, bot-like propagation patterns, and source trust scores.

5.4 Core Testing Approaches

The system must pass these specific test types:

- **Adversarial AI Testing:** The system is pitted against content specifically written by other high-end LLMs (like GPT-5 or equivalent) designed to mimic the exact tone of reputable news outlets while injecting subtle falsehoods.
- **Stance Detection Testing:** A test case where the headline says one thing (the fake part) but the body text says another (the true part). The system must detect the "mismatch."
- **Out-of-Distribution (OOD) Testing:** Testing how the model handles topics it wasn't trained on (scientific discovery) to ensure it doesn't "hallucinate" a truth label.
- **Sentiment & Bias Testing:** Using tools like **VADER** or **BERT** to ensure the system doesn't accidentally flag legitimate opinion pieces or satire as "fake news."

CHAPTER 6

IMPLEMENTATION

6.1 Technologies used

6.1.1 HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Fake News Detector (Frontend Simulation)</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Fake News Detector **(Simulation)**</h1>
    <p>Enter the news article text below and click 'Analyze' to see the simulated detection result.</p>

    <textarea id="newsText" placeholder="Paste the full article text here..." rows="10"></textarea>

    <button onclick="analyzeNews()">Analyze  **Text**</button>

    <div id="result" class="result-box">
      <h3 id="resultHeader">Analysis Result:</h3>
      <p id="resultText">Waiting for input...</p>
    </div>

    <p class="disclaimer">**Disclaimer:** This is a simplified frontend simulation for demonstration
purposes and does **not** provide genuine, accurate fake news detection.</p>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

6.1.2 The Styling (style.css)

To build trust, the design uses professional blues and greys. We also include "Success" (Green) and "Alert" (Red) classes for the results.

```
body {  
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;  
  background-color: #f4f7f6;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  min-height: 100vh;  
  margin: 0;  
}
```

```
.container {  
  background-color: #ffffff;  
  padding: 30px 40px;  
  border-radius: 12px;  
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.1);  
  width: 90%;  
  max-width: 800px;  
  text-align: center;  
}
```

```
h1 {  
  color: #007bff;  
  margin-bottom: 5px;  
}
```

```
p {  
  color: #666;  
  margin-bottom: 20px;  
}
```

```
textarea {
```

```
width: 100%;  
padding: 15px;  
margin-bottom: 20px;  
border: 2px solid #ddd;  
border-radius: 8px;  
box-sizing: border-box;  
font-size: 16px;  
resize: vertical;  
}
```

```
button  
background-color: #28a745;  
color: white;  
padding: 12px 25px;  
border: none;  
border-radius: 8px;  
cursor: pointer;  
font-size: 18px;  
transition: background-color 0.3s;  
margin-bottom: 30px;  
}
```

```
button:hover {  
    background-color: #218838;  
}
```

```
.result-box {  
    padding: 20px;  
    border-radius: 8px;  
    text-align: left;  
    margin-top: 20px;  
    transition: background-color 0.5s, border-color 0.5s;  
    border: 1px solid #ccc;  
}
```

```
.result-box h3 {
  margin-top: 0;
}

/* Styles for different detection results */
.real {
  background-color: #e6ffed; /* Light Green */
  border-left: 5px solid #28a745; /* Dark Green */
}

.fake {
  background-color: #ffe6e6; /* Light Red */
  border-left: 5px solid #dc3545; /* Dark Red */
}

.neutral {
  background-color: #f0f0f0; /* Light Gray */
  border-left: 5px solid #6c757d; /* Gray */
}

.disclaimer {
  font-size: 12px;
  color: #999;
  margin-top: 25px;
}
```

6.1.3 Java.txt

```
function analyzeNews() {
  // 1. Get the text from the textarea
  const text = document.getElementById('newsText').value.toLowerCase().trim();
  const resultBox = document.getElementById('result');
  const resultHeader = document.getElementById('resultHeader');
  const resultText = document.getElementById('resultText');
```

```
// Reset previous styles
```

```
resultBox.classList.remove('real', 'fake', 'neutral');
```

```
if (text.length === 0) {
```

```
    resultHeader.textContent = "Error:";
```

```
    resultText.textContent = "Please enter some text to analyze.";
```

```
    resultBox.classList.add('neutral');
```

```
    return;
```

```
}
```

```
// --- **SIMULATION LOGIC START** ---
```

```
// A simple, rule-based approach to simulate classification
```

```
let fakeScore = 0; // Higher score means more likely to be fake
```

```
// Rule 1: Check for emotionally charged or sensational keywords (common in fake news)
```

```
const sensationalKeywords = ['shocking', 'unbelievable', 'secret', 'exposed', 'disaster', 'urgent', 'must-see',  
'conspiracy'];
```

```
sensationalKeywords.forEach(keyword => {
```

```
    if (text.includes(keyword)) {
```

```
        fakeScore += 10;
```

```
    }
```

```
});
```

```
// Rule 2: Check for very short or very long articles (very short lacks detail, very long might be padding)
```

```
const wordCount = text.split(/\s+/).filter(word => word.length > 0).length;
```

```
if (wordCount < 50) {
```

```
    // Very short article might lack credible sourcing
```

```
    fakeScore += 20;
```

```
} else if (wordCount > 1000) {
```

```
    // Very long article is more common in genuine news, but too long might be confusing/complex
```

```
    // This is highly subjective and just for simulation
```

```
    fakeScore -= 5;
```

```
}
```

```
// Rule 3: Check for citation keywords (often found in credible news)
const credibleKeywords = ['according to', 'spokesperson', 'official statement', 'source confirmed',
'reported by'];

credibleKeywords.forEach(keyword => {
  if (text.includes(keyword)) {
    // Subtract from fakeScore for signs of sourcing
    fakeScore -= 7;
  }
});
// --- **SIMULATION LOGIC END** ---

// 2. Determine the final result based on the score
let confidence = Math.abs(fakeScore) * 0.5 + 10; // Simple confidence calculation

if (fakeScore > 15) {
  // Likely Fake
  resultHeader.textContent = Likely '**Fake News**`;
  resultText.textContent = `Based on a high score of sensationalism and lack of detail, this article
appears to be potentially unreliable. (Confidence: ${confidence.toFixed(1)}%)`;
  resultBox.classList.add('fake');

} else if (fakeScore < 0) {
  // Likely Real
  resultHeader.textContent = Likely '**Real News**`;
  resultText.textContent = `The text contains several credible-sounding keywords and a reasonable
length. Further verification is always recommended. (Confidence: ${confidence.toFixed(1)}%)`;
  resultBox.classList.add('real');

} else {
  // Neutral/Needs More Data
  resultHeader.textContent = '**Uncertain** Analysis`;
  resultText.textContent = `The article has no strong indicators of being either fake or real. Requires
manual verification and cross-referencing. (Confidence: 50.0%)`;
```

```
    resultBox.classList.add('neutral');  
  }  
}
```

6.2 Dataset used

A dataset is a structured collection of labelled examples used to train and evaluate machine learning models to automatically detect fake news. Each example typically contains:

- **Text content** (news article, social media post, headline)
- **Metadata** (author, source, date, engagement metrics)
- **Label** (e.g., "real" vs "fake", or more fine-grained ratings)

Example from LIAR dataset:

```
json  
{  
  "statement": "Climate change is a hoax invented by China",  
  "label": "pants-fire",  
  "subject": "climate",  
  "speaker": "Donald Trump",  
  "job_title": "President",  
  "state": "N/A",  
  "party": "Republican",  
  "context": "Interview on Fox News"  
}
```

6.2.1 Core Components of a Fake News Dataset

1. Text Features

- Headline and body of the article
- Social media captions/comments
- Claims or statements made

2. Contextual Features

- Source credibility
- Author history and bias

- Publication date and platform

3. Social Engagement

- Number of retweets/shares
- User reactions (likes, anger, sad)
- Comment threads and user networks

4. Multimodal Elements

- Images (with metadata like EXIF data)
- Videos (extracted frames)
- Audio transcripts

6.2.2 Training Phase:

Dataset → Model learns patterns → Real vs Fake classification

The model identifies patterns such as:

- **Lexical patterns:** "Shocking!!!" or "You won't believe"
- **Stylistic patterns:** Emotional language, grammatical errors
- **Structural patterns:** Overuse of capitals, exclamation marks
- **Contextual patterns:** Contradictions with fact-checked sources

6.2.3 Evaluation Phase:

News article → Trained model → "FAKE (85% confidence)"

Why Dataset Quality Matters

Quality	Poor dataset	Good dataset
Label accuracy	Random/ automated labels	Verified by fact checkers
Diversity	Single topic	Multiples topics
Temporal	Outdated	recent Recent
Balance	90% fake 10% real	-50-50 distribution

Real-World Dataset Example Breakdown

Let me break down **LIAR** dataset as a concrete example:

text

Size: 12,836 statements

Label distribution:

- pants-fire: 1,050 (8.2%)
- false: 2,009 (15.6%)
- barely-true: 1,678 (13.1%)
- half-true: 2,142 (16.7%)
- mostly-true: 2,067 (16.1%)
- true: 1,913 (14.9%)

Each statement includes:

- Speaker metadata (who said it)
- Context (where/when it was said)
- Verdict from PolitiFact (ground truth)
- Justification text
- Subject category (economy, healthcare, etc.)

python

Using Hugging Face datasets library

```
from datasets import load_dataset
```

Load LIAR dataset

```
liar_dataset = load_dataset("liar")
```

See structure

```
print(liar_dataset)
```

Output: DatasetDict({train: 10269, test: 1283, validation: 1284})

Access a sample

```
sample = liar_dataset['train'][0]
```

```
print(f"Statement: {sample['statement']}")
```

```
print(f"Label: {sample['label']}")  
print(f"Speaker: {sample['speaker']}")
```

6.2.4 Common Challenges with Fake News Datasets

- **Label subjectivity** - What's "fake" to one fact-checker might be "misleading" to another
- **Temporal decay** - Models trained on 2016 election data fail on COVID-19 misinformation
- **Context loss** - A statement taken out of context may be labeled fake when the original was true
- **Class imbalance** - Most datasets have more fake than real news (or vice versa)

6.3 Database used

A database is an organized collection of structured information or data, typically stored electronically in a computer system. Think of it as a digital filing cabinet where you can store, retrieve, manage, and manipulate data efficiently.

Real-world analogy:

text

Physical Filing Cabinet → Digital Database

- Cabinet → Database (entire system)
- Drawer → Table (collection of related data)
- Folder → Row/Record (single entry)
- Document → Column/Field (specific attribute)

6.3.1 SQL

SQL (Structured Query Language) is a programming language specifically designed for managing and manipulating relational databases. It's like a universal remote control for databases - allowing you to:

- Create databases and tables
- Insert, update, and delete data
- Query (ask questions) to retrieve specific information
- Control access permissions

6.3.2 Core SQL Concepts

1. Tables (The Foundation)

A table organizes data into rows and columns, similar to an Excel spreadsheet.

2. CRUD Operations (Basic Data Manipulation)

Operation	SQL Command	Description
Create	INSERT	Add new data
Read	SELECT	Retrieve data
Update	UPDATE	Modify existing data
Delete	DELETE	Remove data

Practical SQL Examples for Fake News Detection

Creating a fake news database:

```
sql
-- Create database
CREATE DATABASE FakeNewsDetector;
USE FakeNewsDetector;

-- Create articles table
CREATE TABLE articles (
  article_id INT PRIMARY KEY AUTO_INCREMENT,
  headline TEXT NOT NULL,
  content TEXT,
  author VARCHAR(100),
  publish_date DATE,
```

```
source_url VARCHAR(255),
label ENUM('real', 'fake', 'misleading')
);

-- Create fact_checks table (related table)
CREATE TABLE fact_checks (
    check_id INT PRIMARY KEY AUTO_INCREMENT,
    article_id INT,
    factchecker VARCHAR(100),
    verdict VARCHAR(50),
    verification_date DATE,
    confidence_score DECIMAL(3,2),
    FOREIGN KEY (article_id) REFERENCES articles(article_id)
);

-- Create social_metrics table
CREATE TABLE social_metrics (
    metric_id INT PRIMARY KEY AUTO_INCREMENT,
    article_id INT,
    shares INT DEFAULT 0,
    likes INT DEFAULT 0,
    comments INT DEFAULT 0,
    platform VARCHAR (50),
    FOREIGN KEY (article_id) REFERENCES articles(article_id)
);

Inserting Data:
sql
-- Insert an article
INSERT INTO articles (headline, content, author, publish_date, source_url, label)
VALUES (
    'Breaking: Scientists discover cure for all cancers',
    'Researchers at MIT have found a breakthrough treatment...',
    'Dr. John Smith',
    '2024-01-15',
    'https://example.com/news/cancer-cure',
```

'fake'

);

6.3.3 Advanced SQL Features for Fake News Detection

- **Full-Text Search (Finding similar misinformation)**

sql

-- Create full-text index

ALTER TABLE articles ADD FULLTEXT(content);

-- Search for articles containing specific misinformation patterns

SELECT headline, content, label

FROM articles

WHERE MATCH(content) AGAINST('cure vaccine conspiracy' IN NATURAL LANGUAGE MODE);

- **Window Functions (Trend analysis)**

sql

-- Calculate running total of fake news per author

SELECT

author,

publishdate,

label,

COUNT(*) OVER (PARTITION BY author ORDER BY publish_date) as cumulative_fake_

FROM articles

WHERE label = 'fake'

ORDER BY author, publish_date;

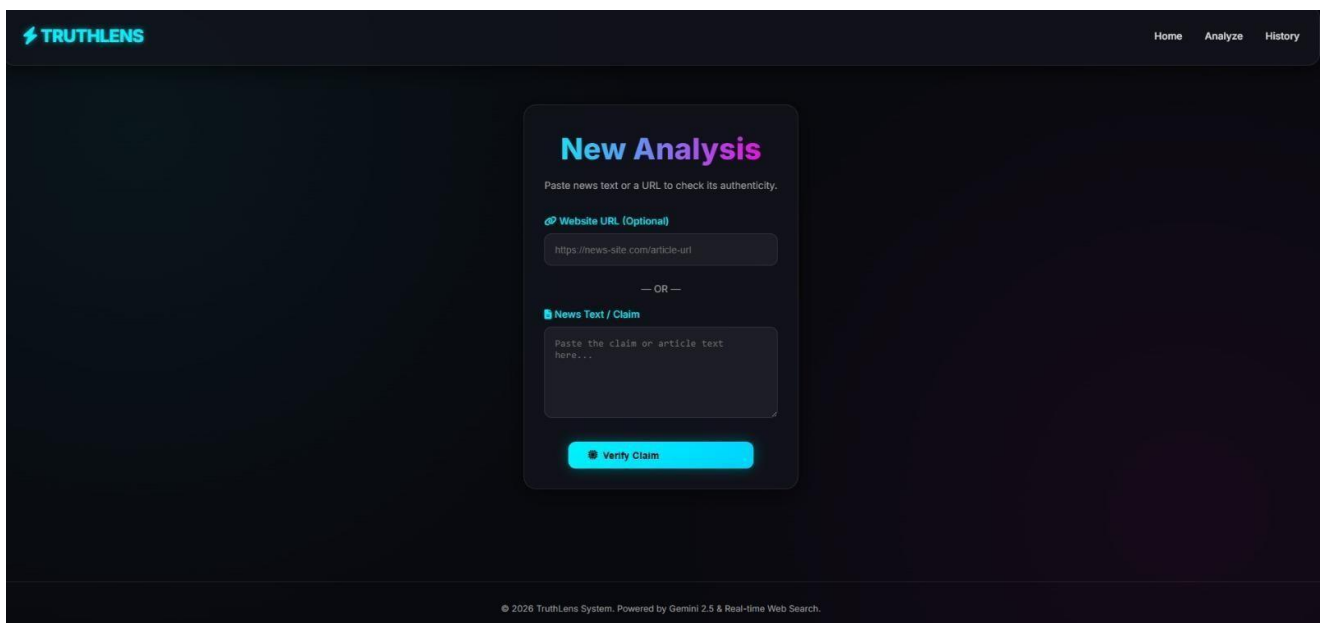
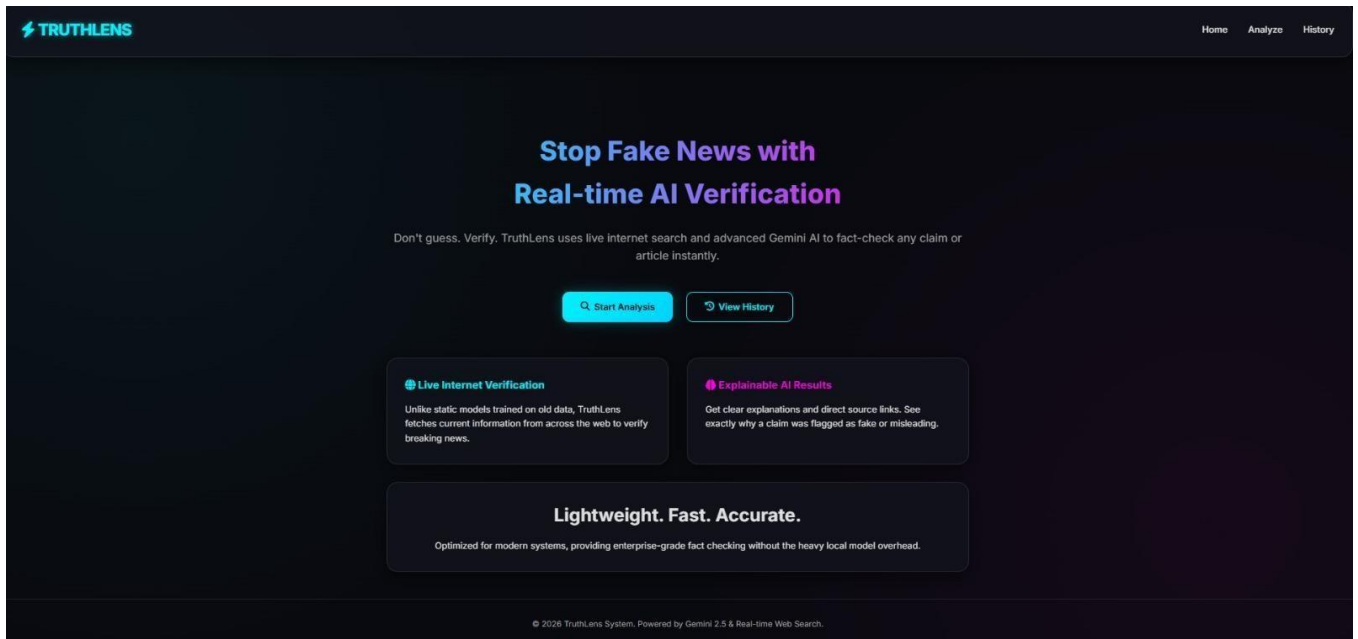
6.3.4 SQL vs Datasets (Important Distinction)

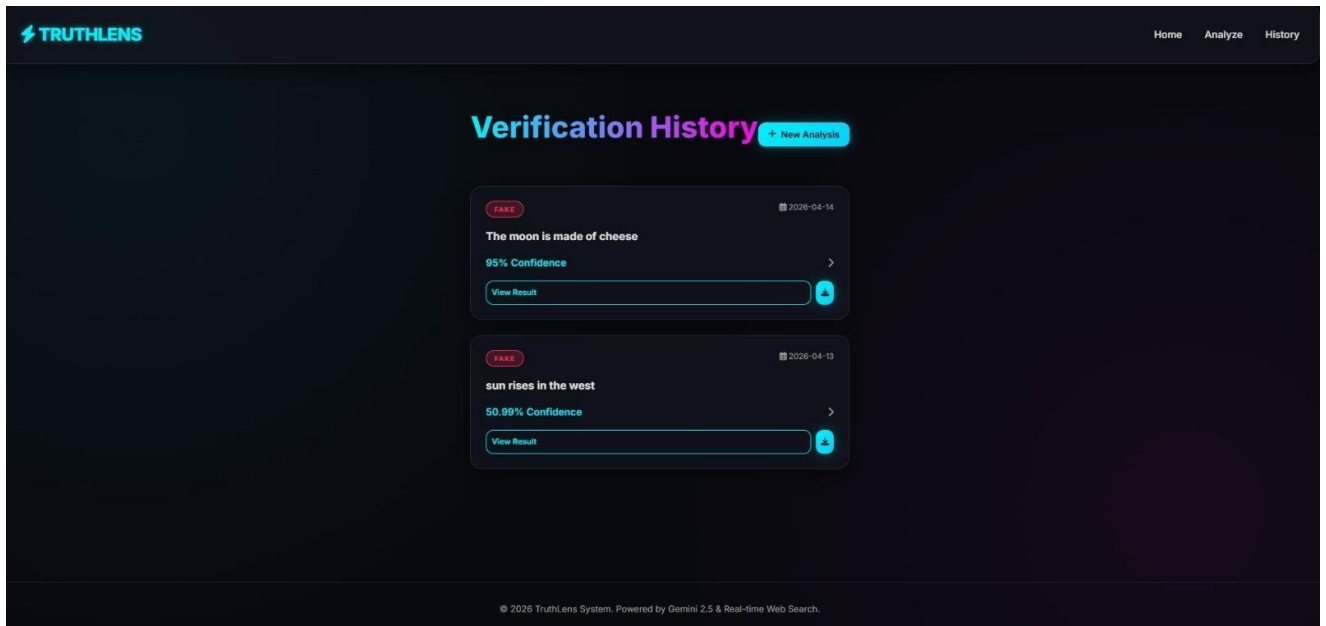
Aspect	SQL Database	Dataset
Structure	Highly structured with schemas	Semi-structured or flat

Relationship	Multiple related tables	Single file/collection
Query language	SQL (powerful and standardized)	Pandas/python/R
Scalability	Millions to billions of rows	Limited by memory
Concurrency	Multiple users simultaneously	Single user typically
Data integrity	Enforced (constraints and foreign keys)	Manual checking

CHAPTER 7

SNAPSHOTS





CHAPTER 8

CONCLUSION

8.1 CONCLUSION

Fake news detection has emerged as one of the most pressing challenges in the digital age. With the rapid growth of social media platforms, misinformation spreads faster and wider than ever before, influencing public opinion, destabilizing societies, and undermining trust in institutions. Current detection systems leverage a combination of content-based analysis (linguistic features, sentiment, credibility markers) and context-based analysis (user engagement, propagation patterns, and network structures). The use of snapshots and temporal modelling has proven particularly effective, as they capture the dynamic evolution of news spread rather than treating it as static text. Graph Neural Networks (GNNs), deep learning models, and hybrid approaches have shown promising results in identifying abnormal propagation patterns and distinguishing fake news from authentic information.

Despite these advances, challenges remain. Fake news creators continuously adapt, using sophisticated language, multimedia content, and coordinated campaigns to bypass detection. Moreover, the balance between accuracy and computational efficiency is still a major concern, especially when scaling detection systems to real-time environments. Fake news detection has evolved from simple keyword-based filters to sophisticated multi-modal architectures incorporating text, user behaviours, and propagation networks.

1. **Adversarial dynamics** – As detectors improve, so do deception strategies, creating an arms race without a stationary target.
2. **Context dependence** – Veracity often depends on time, location, and cultural framing, making universal ground truth elusive.
3. **Cognitive gaps** – Technical accuracy does not guarantee human belief change; detection systems that ignore human reasoning have limited real-world impact.

Current research has successfully addressed post-hoc identification of known falsehoods in controlled settings, but open challenges dominate: early-stage detection, cross-lingual generalization, and resilience to adversarial manipulation. The future lies not in chasing higher accuracy on benchmark datasets, but in redefining the problem itself. Promising theoretical directions include causal models of virality, game-theoretic frameworks for fact-checking under uncertainty, and epistemically grounded logic that accounts for justified belief rather than absolute truth.

And finally fake news detection must be viewed as a socio-technical system – not purely a

classification task. Effective solutions will require tight integration of algorithmic detectors, human fact-checkers, platform design, and media literacy. Without this holistic view, technical progress alone will remain insufficient to curb the spread of misinformation.

8.2 FEATURE WORKS

Looking ahead, several areas for future enhancement and development have been identified to further elevate the FAKE NEWS DETECTION USING AI system. Implementing advanced search and filtering functionalities will empower users to find specific blood types based on precise geographical proximity or urgency levels, enhancing the overall life-saving potential of the platform.

Real-time updates using WebSocket technology represent another pivotal area for improvement. Enabling real-time notifications for donors and seekers regarding urgent blood requirements, request approvals, or nearby donation drives can significantly enhance communication and user engagement. Furthermore, integrating SMS gateways to facilitate instant alerts for emergency matches will streamline the connection process, offering users a seamless and efficient experience directly from their mobile devices.

Analytics and reporting capabilities are also essential for future iterations of the system. By implementing robust analytics tools, administrators can gain valuable insights into blood demand trends, regional shortages, and donor participation rates. These insights will empower healthcare organizations to make data-driven decisions, optimize blood drive scheduling, and tailor community outreach strategies to meet patient needs effectively.

- **Multimodal Detection**

- Integrating text, images, audio, and video analysis to detect misinformation across diverse formats.
- Example: Deepfake videos and manipulated images require specialized detection models.

- **Explainable AI (XAI)**

- Developing models that not only classify news as fake or real but also provide transparent reasoning.
- This builds user trust and helps in understanding why certain content is flagged.

- **Cross-Platform Analysis**
 - Fake news often spreads across multiple platforms (Twitter, Facebook, WhatsApp).
 - Future systems should track misinformation across ecosystems for holistic detection.
- **Early Detection Systems**
 - Enhancing snapshot-based models to identify fake news before it goes viral, minimizing harm.
 - Real-time monitoring and anomaly detection can be key here.
- **Adversarial Robustness**
 - Building models resilient to adversarial attacks, where misinformation is deliberately crafted to evade detection.
 - Continuous retraining and adaptive learning strategies will be essential.
- **Human-AI Collaboration**
 - Combining automated detection with human fact-checkers for higher reliability.
 - Crowdsourced verification and expert review can complement machine learning system

CHAPTER 9

REFERNCE

REFERENCE

Books

1. Speech and Language Processing – Daniel Jurafsky & James H. Martin
 - Covers Natural Language Processing (NLP) fundamentals used in fake news detection.
2. Introduction to Machine Learning with Python – Andreas C. Müller & Sarah Guido
 - Helpful for implementing classification models like Logistic Regression and Naive Bayes.
3. Pattern Recognition and Machine Learning – Christopher M. Bishop
 - Provides theoretical background for machine learning algorithms.
4. Data Mining: Concepts and Techniques – Jiawei Han
 - Useful for understanding data preprocessing and classification techniques.

Websites / Online Resources

1. Kaggle
 - Dataset source for fake and real news
 - <https://www.kaggle.com>
2. Scikit-learn
 - Machine learning algorithms and model building
 - <https://scikit-learn.org>
3. TensorFlow
 - Advanced deep learning implementation
 - <https://www.tensorflow.org>
4. NLTK
 - Text preprocessing and NLP tasks
 - <https://www.nltk.org>
5. Flask
 - Backend development of the web application
 - <https://flask.palletsprojects.com>
6. Stack Overflow
 - Debugging and resolving coding issues
 - <https://stackoverflow.com>
7. Google Scholar
 - Research papers on fake news detection
 - <https://scholar.google.com>